Reg. No										

St. Joseph's College of Arts & Science (Autonomous)
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AMCS101S – ALLIED MATHEMATICS - I

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. If α, β, γ are the roots of $2x^3 + 3x^2 + 5x + 6 = 0$ find $\sum\alpha$, $\sum\alpha\beta$ & $\alpha\beta\gamma$.

2. Multiply the roots of $x^3 - 3x + 1 = 0$ by 10.

3. State cayley-Hamilton theorem.

4. Expand $\tan n\theta$.

5. Find the n^{th} derivative of $y = e^{ax}$.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Form the equation one of whose roots is $\sqrt{5} + \sqrt{3}$.

7. Solve: $x^4 + 10x^3 + 26x^2 - 10x + 1 = 0$.

8. Find the rank of Matrix, $A = \begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix}$.

9. Find θ approximately to the nearest minute of $\cos\theta = \frac{1681}{1682}$.

10. If $x = r\cos\theta$, y=rsinθ find $\frac{\partial(x,y)}{\partial(r,\theta)}$.

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) Solve the equation $27x^3 + 42x^2 - 25x - 8 = 0$ whose roots are in GP.

(or)

b) If α, β, γ are the roots of the equation, $x^3 + px^2 + qx + r = 0$ find the value of i) $\sum \alpha^2$ ii) $\sum 1/\alpha$ iii) $\sum 1/\alpha\beta$ iv) $\sum \alpha^2\beta^2$

12. a) Solve the equation, $6x^3 - 11x^2 - 3x + 2 = 0$ given that, its roots are in HP.

(or)

b) Find the real root of $3x - \cos x - 1 = 0$ by Newton – Raphson method.

13. a) Find the Eigen values and Eigen vectors of the matrix: $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$.

(or)

b) If the matrix A is given by, $A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 3 & 2 & -1 \end{pmatrix}$ obtain a matrix B such that, $AB = BA = I$.

14. a) Prove that, $\cos^6\theta = \frac{1}{32}\left(\cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10\right)$

(or)

b) Find i) $\lim_{x\to 0}\frac{x-\sin x}{x^3}$ ii)Separate the real & Imaginary parts of $\log\left(1+i\right)^{1-i}$.

15. a) Find the value of the Jacobian $\frac{\partial(u,v)}{\partial(r,\theta)}$ where $u = x^2 - y^2, v = 2xy$ and $x = r\cos\theta, y = r\sin\theta$.

(or)

b) Find the radius of cusvature for the curve $\sqrt{x} + \sqrt{y} = 1$ at $\left(1/4, 1/4\right)$.

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